

Utkarsh Shaning

LinkedIn | GitHub | Portfolio | LeetCode

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EDUCATION

- **Jaypee University of Information Technology** Solan, Himachal Pradesh
Bachelor of Technology in Computer Science; CGPA: 7.79 2024 – 2028
 - Relevant Coursework: Database Management Systems (DBMS), Object-Oriented Programming (OOPS), Data Structures and Algorithms

PROJECTS

- **Distributed Task Queue System** *GitHub*
Tech Stack: Java, Spring Boot, RabbitMQ, Docker
 - Built a high-throughput, distributed message queue capable of handling asynchronous background tasks efficiently.
 - Implemented load-balancing algorithms to distribute workloads across multiple worker nodes, ensuring system stability under heavy traffic.
- **Scalable Microservices-Based Salon Booking Platform** *GitHub*
Tech Stack: Spring Boot, RabbitMQ, React.js, MongoDB, API Gateway, Stripe, Keycloak, Docker
 - Designed RESTful microservices using layered architecture (Controller–Service–Repository) ensuring modularity and separation of concerns.
 - Implemented service discovery with Eureka Server and centralized routing and filtering via API Gateway.
 - Developed asynchronous event-driven workflows using RabbitMQ to decouple booking, notification, and payment services.
 - Integrated Stripe webhooks for real-time transaction handling and synchronized MongoDB updates across services.
 - Secured distributed services using Keycloak with JWT-based authentication and RBAC.
 - Enabled resilient inter-service communication using OpenFeign clients with circuit breaker mechanisms.
 - Containerized all services using Docker Compose for local orchestration and deployment.
- **Retrieval-Augmented Generation (RAG) Question Answering System** *GitHub | Live*
Tech Stack: FastAPI, LangChain, ChromaDB, Python, Streamlit
 - Developed RAG-based architecture for contextual question answering over unstructured PDF documents.
 - Implemented document chunking, embedding generation, and semantic similarity search using LangChain and ChromaDB.
 - Exposed inference through FastAPI REST endpoints and built UI for document ingestion and conversational querying.

TECHNICAL SKILLS

- **Languages:** Java, Python
Frameworks: Spring Boot, Spring MVC, FastAPI, LangChain
Databases: MongoDB, MySQL
Messaging & Distributed Systems: RabbitMQ
Security: Keycloak, JWT, RBAC, Spring Security
Dev Tools: Docker, Postman, Swagger/OpenAPI, Git, Streamlit

HONORS & AWARDS

- Won **3rd Position (Innovation Award)** at **Cyber AI Hackathon 2025** (International – University of Derby, UK, supported by American Society of Engineers); competed among 100+ teams, drove backend development and AI logic for the winning solution. *Live | GitHub*